

已知函数

$$f(x) = |x|$$

的傅里叶级数前 n 项为 $S_n(x)$

求

$$T_n = \frac{1}{2\pi} \int_0^{2\pi} [f(x) - S_n(x)]^2 dx$$
$$\lim_{n \rightarrow \infty} T_n$$

先求出目标 $Fourier$ 级数

$$S_n(x) = \frac{\pi}{2} + \sum_{k=1}^n \frac{2}{\pi} \frac{-1 + (-1)^k}{k^2} \cos(kx)$$

$$S_n(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \frac{1}{(2k+1)^2} \cos(2k+1)x$$

则要求的数列为

$$\begin{aligned} & \frac{1}{2\pi} \int_0^{2\pi} \frac{16}{\pi^2} \left[\sum_{k=\lfloor \frac{n+1}{2} \rfloor}^{\infty} \frac{1}{(2k+1)^2} \cos(2k+1)x \right]^2 dx \\ &= \frac{8}{\pi^3} \sum_{k=\lfloor \frac{n+1}{2} \rfloor}^{\infty} \frac{1}{(2k+1)^2} \int_0^{2\pi} \cos^2(2k+1)x dx \\ &= \frac{8}{\pi^2} \left[\beta(4) - \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \frac{1}{(2k+1)^2} \right] \\ \lim_{n \rightarrow \infty} &= \frac{8}{\pi^2} \left[\beta(4) - \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \frac{1}{(2k+1)^2} \right] = 0 \end{aligned}$$